

WhatsApp & Email Tracker Bot - Ubuntu Deployment Guide

Overview

This guide will help you deploy the WhatsApp & Email Tracker Bot on an Ubuntu server step by step.

1. Get the Project

Clone the project from GitHub and navigate to the project folder:

```
bash

git clone https://github.com/huzaifasaeed123/WhatsApp_bot.git
cd "WhatsApp_bot/WhatsApp & Email Tracker Bot"
```

2. Install Node.js

Check if Node.js is installed:

```
bash

node -v
```

If not installed, install Node.js (recommended v18+):

```
bash

curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs
```

Verify installation:

```
bash

node -v
npm -v
```

3. Install Project Dependencies

Inside the project directory, run:

```
bash
```

```
npm install
```

This installs all required dependencies from package.json.

4. Create .env File

Create a .env file using the following command:

```
bash
```

```
nano .env
```

Paste the following content and update only the highlighted values:

```
env
```

```
OPENAI_API_KEY=YOUR_OPENAI_API_KEY  
EMAIL_USER=your_email@gmail.com  
EMAIL_PASS=your_gmail_app_password  
EMAIL_HOST=imap.gmail.com  
EMAIL_PORT=993  
MONGODB_URI=mongodb://localhost:27017/monitor_bot  
PORT=3000  
NODE_ENV=development
```

Important Notes:

- **EMAIL_PASS** should be a Gmail App Password, not your normal Gmail password
- **OPENAI_API_KEY** should be your OpenAI API key
- Keep other values as default unless needed to change

Save the file:

Press `Ctrl+O`, then `Enter`, then `Ctrl+X` to exit.

5. Install MongoDB

Install MongoDB on Ubuntu:

```
bash
```

```
sudo apt update
sudo apt install -y mongodb
sudo systemctl start mongodb
sudo systemctl enable mongodb
```

Check MongoDB status:

```
bash

sudo systemctl status mongodb
```

Ensure MongoDB is running before starting the bot.

6. Install PM2

PM2 is used to run the bot in the background:

```
bash

sudo npm install -g pm2
pm2 -v
```

7. Start the Bot with PM2

Start the bot using PM2:

```
bash

pm2 start server.js --name "WhatsApp_Email_Bot"
```

Check bot logs:

```
bash

pm2 logs WhatsApp_Email_Bot
```

8. Set PM2 to Auto-Start on Reboot

```
bash

pm2 startup systemd
pm2 save
```

The bot will now automatically restart if the server reboots.

9. Access the Bot

- The bot will run on **PORT 3000** by default (as set in .env)
- Ensure your server firewall allows the port if needed:

```
bash
sudo ufw allow 3000
sudo ufw status
```

10. Quick Command Summary

For quick deployment, run these commands in sequence:

```
bash
git clone https://github.com/huzaifasaeed123/WhatsApp_bot.git
cd "WhatsApp_bot/WhatsApp & Email Tracker Bot"
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs
npm install
nano .env
sudo apt install -y mongodb
sudo systemctl start mongodb
sudo systemctl enable mongodb
sudo npm install -g pm2
pm2 start server.js --name "WhatsApp_Email_Bot"
pm2 startup systemd
pm2 save
pm2 logs WhatsApp_Email_Bot
sudo ufw allow 3000
```

11. Post-Deployment Verification

Access the application:

Open your web browser and go to:

```
http://YOUR_SERVER_IP:3000
```

Verify services are running:

```
bash
pm2 status
sudo systemctl status mongod
```

12. Troubleshooting

If the bot doesn't start:

1. Check logs: `pm2 logs WhatsApp_Email_Bot`
2. Verify .env file: `cat .env`
3. Check MongoDB: `sudo systemctl status mongod`
4. Restart services: `pm2 restart WhatsApp_Email_Bot`

If you can't access the web interface:

1. Check firewall: `sudo ufw status`
 2. Verify port: `netstat -tulpn | grep :3000`
 3. Check server logs: `pm2 logs WhatsApp_Email_Bot`
-

13. Maintenance Commands

View application status:

```
bash
pm2 status
```

Restart the application:

```
bash
pm2 restart WhatsApp_Email_Bot
```

Stop the application:

```
bash
pm2 stop WhatsApp_Email_Bot
```

View real-time logs:

```
bash
```

```
pm2 logs WhatsApp_Email_Bot --lines 100
```

14. Required Credentials Setup

OpenAI API Key:

1. Go to: <https://platform.openai.com/api-keys>
2. Login to your OpenAI account
3. Click "Create new secret key"
4. Copy the key (starts with `sk-proj-`)

Gmail App Password:

1. Go to: <https://myaccount.google.com/apppasswords>
 2. Select app: "Mail"
 3. Select device: "Other (custom name)"
 4. Generate password
 5. Copy the 16-character password (with spaces)
-

15. Security Considerations

- Keep your `.env` file secure and never commit it to version control
 - Use strong passwords for email accounts
 - Consider using environment variables on production servers
 - Regularly update dependencies: `npm update`
 - Monitor logs for any suspicious activity
-

16. Features Overview

WhatsApp Integration:

- QR code-based authentication
- Real-time group monitoring

- Message tracking and processing
- Automated responses to specific shipments

Email Integration:

- IMAP-based email monitoring
- Automatic email processing
- Support for Gmail and other providers
- Email response capabilities

AI Processing:

- OpenAI GPT integration for logistics data extraction
- Automatic extraction of shipping details
- Support for multiple languages and formats
- Structured data output

Web Dashboard:

- Real-time message monitoring
- Advanced search and filtering
- Export to Excel functionality
- Service status monitoring

Your WhatsApp & Email Tracker Bot is now successfully deployed and running on Ubuntu!

For support or issues, please check the troubleshooting section or contact the development team.